

Advanced Algorithms

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Autumn 2026

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1 Exercise Set 1

Problem 1.1.

$$U, \quad |U| = n, \quad f : 2^U \rightarrow \mathbb{R}_{\geq 0}.$$

$$\text{monotone : } \forall S \subseteq U, \forall T \subseteq U, S \subseteq T : f(S) \leq f(T).$$

$$\text{submodular : } \forall i \in U, \forall S \subseteq U, \forall T \subseteq U, S \subseteq T : f(S \cup \{i\}) - f(S) \geq f(T \cup \{i\}) - f(T).$$

Solution. □

Problem 1.2.

Solution. □

Problem 1.3.

Solution. □

2 Exercise Set 2

Problem 2.1 (Rounding for Bin Packing).

Solution. □

Problem 2.2 (Target shooting).

Solution. □

Problem 2.3 (Counting satisfying assignments).

Solution.

□

Problem 2.4 (Cut Counting).

Solution.

□